

GLASGOW COLLEGE UESTC

Exam paper

Electromagnetic Field and Microwave Technology (4002)

Date: July 1st 2024

Time: 09:30-11:30

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

For Multiple Choice Questions, use the dedicated answer sheet provided.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

DATA/FORMULAE SHEET IS PROVIDED AT THE END OF PAPER

- Q1 (a) State the integral form and the differential form of Maxwell's equations, and explain the difference between them. [4]
- (b) Show the basic equations for the steady electric field; derive the boundary condition of the steady electric field on the interface using correct models. [5]
- (c) A dipole antenna generates electromagnetic waves in air, the magnetic field intensity of this wave is $\vec{H} = \vec{e}_x 0.15 \cos(9.36 \times 10^8 t - 3.12 y)$ A/m, determine the magnitude of the displacement current density. [6]
- (d) Illustrate the three types of the polarization for electromagnetic waves; determine the type of polarization for the following uniform plane wave

$$\vec{E} = \vec{e}_x E_m \sin(\omega t + kz - \frac{\pi}{4}) + \vec{e}_y E_m \cos(\omega t + kz - \frac{\pi}{4})$$
 [6]
- Q2 (a) What is the phase difference between $\vec{E}$ and $\vec{H}$ for a uniform plane wave propagates in perfect dielectric and a good conductor? [4]
- (b) Explain the difference between travelling wave, standing wave and travelling-standing wave. [6]
- (c) There is an infinite cylinder with permeability of μ and radius of a , and outside the cylinder is air (μ_0). A line current I exist in the axis of the cylinder, as shown in **Figure Q2-1**. Find the distribution magnetization vector M and the surface magnetized current density. [7]
- (d) Illustrate the mutual inductance between a long straight wire and a rectangular loop, the wire is at the interface of air (μ_0) and magnetic material (μ), while the loop is inside the magnetic material (μ), as shown in **Figure Q2-2**. [8]

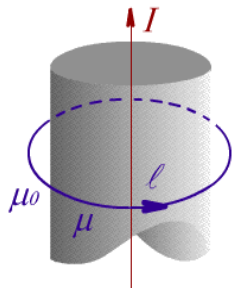


Figure Q2-1

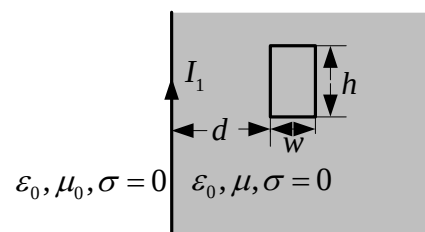


Figure Q2-2

Continued overleaf

Q3

In time-varying field, the electric field and magnetic field can be induced by each other. The phasor expression of the electric field intensity for plane wave

in air is $\vec{E}(r) = \vec{e}_x 10^{-3} e^{-j10\pi z} + \vec{e}_y 10^{-3} e^{-j(10\pi z - \frac{\pi}{2})}$ V / m V/m.

- (a) Find the propagation direction $\vec{e}_n$ and the frequency of the plane wave. [6]
- (b) Determine the polarization of the plane wave. [6]
- (c) Find the phasor expression for the magnetic field intensity. [7]
- (d) State the average Poynting vector. [8]

Q4

A uniform plane wave, with frequency of 6GHz and the amplitude of electric field intensity $E_{im}=10\text{V/m}$, perpendicularly incidents from air to a perfect dielectric ($\sigma=0, \epsilon_r=4, \mu_0$) along +z axis, as shown in **Figure Q4** :

- (a) Find angular frequency and the wavelength in perfect dielectric of the uniform plane wave; [4]
- (b) Explain the phase constant in air β_0 and the instantaneous express of the incident wave $\vec{E}_i(r, t)$. [6]
- (c) Describe the phase constant in perfect dielectric β_1 , the phasor express for the reflected wave $\vec{E}_r(r)$ and the transmitted wave $\vec{E}_t(r)$. [10]
- (d) Show the percentage of energy that is reflected on the interface. Explain how to decrease the reflected energy with one example. [7]

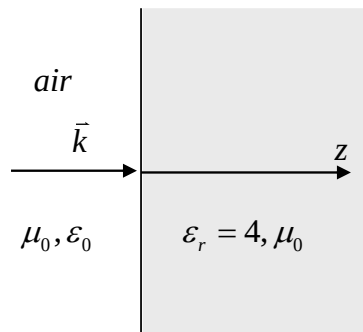


Figure Q4

Continued overleaf

Appendix

Cylindrical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{e}_\rho & \rho \mathbf{e}_\phi & \mathbf{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

Spherical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}$$

End of question paper